



Seizure

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Outlines

- ❖ Definition of the seizure
- ❖ Pathophysiology of the seizure
- ❖ Risk factors of the seizure
- ❖ Clinical Presentation of the seizure
- ❖ Complications of the seizure
- ❖ Diagnostic Tests of the seizure
- ❖ Management of the seizure
- ❖ Nursing management for epilepsy

Pre test

Mention risk
factors of
seizures

Mention clinical
manifestation
of seizures

Definition:

seizure disorder is a temporary, abnormal, sudden, excessive, uncontrolled electrical discharge of neurons of the cerebral cortex.

Status epilepticus (SE), which denotes continuous seizure activity (lasting >5 min or two or more sequential seizures), is a medical emergency

Pathophysiology

Repetitive depolarization
of hyperactive
hypersensitive brain cells

→ abnormal electrical
activity in the brain.

Risk factors for seizure disorder include:

- Epilepsy

- Drug or alcohol abuse

- Drug toxicity (aminophylline)

- Recent head injury

- Infection

- Headache

- Acute metabolic disturbances (hypoglycemia, hyponatremia, hypocalcemia, renal failure)

■ CVA

■ CNS infection (meningitis, encephalitis) CNS trauma or tumor

■ Hypoxemia

■ Fever (children)

■ HTN

■ Allergic reaction

■ Eclampsia related to pregnancy

Clinical Presentation

- From simple staring to prolonged convulsions
- Brief loss of memory, sparkling or flashes, and sensing of an unpleasant odor
- Classified as motor, sensory, autonomic, emotional, or cognitive
- Aura occurring before the seizure, along with tachycardia

■ Alteration in mental state;
confusion or dazed state

■ Automatism such as lip
smacking or chewing

■ Tonic or clonic movements

■ Loss of consciousness or inability
to respond to external stimuli

Complications

- Pulmonary edema
- Pulmonary aspiration
- Cardiac dysrhythmias
- HTN or hypotension
- Hyperthermia

■ Hyperglycemia/hypoglycemia

■ Hypoxia

■ Dehydration and metabolic imbalances

■ Myoglobinuria

■ Oral or musculoskeletal injuries

■ Brain death

Diagnostic Tests

- EEG

- CT scan, MRI, or PET to rule out cerebral lesions

- Serum drug screen to rule out drug or alcohol intoxication

- Serum electrolytes, BUN, calcium, magnesium, glucose

- CBC

- ECG to detect cardiac arrhythmias

- ABGs or pulse oximetry

Management

The primary aim of seizure management is to protect the airway and prevent injury to the patient.

- Administer fast-acting anticonvulsants:

- Lorazepam (Ativan): 0.1 mg/kg up to 4 mg per dose IV. Administer no faster than 2 mg/min. May repeat in 5–10 min.

- Midazolam: 0.2 mg/kg IM up to 10 mg maximum.

- Diazepam (Valium): 0.15 mg/kg IV up to 10 mg per dose Administer no faster than 5 mg/min. May repeat in 5 min.

■ Administer long-acting anticonvulsants:

■ Phenytoin (Dilantin): 20 mg/kg IV.
Administer no faster than 50 mg/min.

■ Phenobarbital (Luminal): 20 mg/kg IV.
Administer no faster than 50–100 mg/min.
May give an additional 5–10 mg/kg 10 min
after loading infusion.

Nursing management for epilepsy

- Identify precipitating factors and preceding aura.

- What was the patient doing at the time of the seizure?

- Where there any unusual symptoms or behaviors before the seizure (e.g., movements, sensations, sounds, tastes, smells)?

- Did the patient know he or she was going to have a seizure?

- **During seizure:**

- Observe seizure type, point of origin, and spread of seizure activity.

- Note length of time of the seizure.

- Note automatisms, such as lip smacking and repeated swallowing.

- Assess LOC, bowel and bladder incontinence, and tongue biting.

- Avoid restraining the patient.

- Avoid forcing airway into patient's mouth when jaws clenched.

- Avoid use of tongue blade.

- Maintain patent airway during seizure.

- **During postictal state (after seizure):**

- Assess vital signs closely; provide ECG monitoring.

- Monitor oxygenation and respiratory status (ABGs, SpO₂, breath sounds).

Turn patient to side-lying position; administer O2 therapy; suction prn.

- Check level of orientation and ability to speak (patient usually sleeps afterward).

- Note headache and signs of increased ICP.

- Check pupil size, eye deviations, and response to auditory and tactile stimuli.

- Note paralysis or weakness of arms or legs.

- Keep oral or nasal airway or ETT at bedside.

- Ensure patient's safety (pad side rails, bed at lowest position).

Post test



MENTION RISK FACTORS OF
SEIZURES



MENTION CLINICAL
MANIFESTATION OF SEIZURES



ANY QUESTION



THANK YOU